Palettes, contrast, and dark mode

Week 4, Lecture 2 · Design for Builders: Ship Beautiful Products as a Founder

Autumn 2026 · Founders, indie hackers, engineers

What we'll cover today

- **Tailwind-style ramps:** how the 50-950 scale is structured and why
- **WCAG contrast:** the numbers, the formula, and the common failure modes
- **Semantic color tokens:** naming colors by function, not by shade
- **Dark mode:** a parallel semantic system, not an inversion

Tailwind-style 9-step ramps

The Tailwind color scale

Tailwind's default palette runs from 50 (lightest) to 950 (darkest) for each color family.

```
Slate/50:   #f8fafc   -- near-white, tinted surface
Slate/100:  #f1f5f9
Slate/200:  #e2e8f0
Slate/300:  #cbd5e1
Slate/400:  #94a3b8
Slate/500:  #64748b   -- mid-gray base
Slate/600:  #475569
Slate/700:  #334155
Slate/800:  #1e293b
Slate/900:  #0f172a   -- near-black foreground
Slate/950:  #020617
```

(Tailwind Labs, 2024)

Why this structure works

Each step is a color with a specific job:

- **50-100:** page background, tinted card background
- **200-300:** borders, dividers, table separators
- **400-500:** placeholder text, disabled element color
- **600:** secondary text (check contrast)
- **700-800:** primary text on light surfaces
- **900-950:** maximum-emphasis foreground

You don't need all 11 steps in every design. You need enough steps to cover the jobs.

Building your ramp: start at 500

Choose your brand color as the 500 base. Then:

- **Step 100:** how light can this color go and still be distinct from white? That's your 100.
- **Step 900:** how dark can this color go before it reads as near-black? That's your 900.
- **Fill outward from 500.** 400 goes between 500 and 300; 600 goes between 500 and 700.

Check: do adjacent steps look meaningfully different? Do any two steps look identical?

WCAG contrast

The contrast ratio formula

WCAG contrast is a ratio between the relative luminance of two colors.

```
Contrast ratio = (L1 + 0.05) / (L2 + 0.05)

where L1 is the lighter color's relative luminance
and L2 is the darker color's relative luminance.

Range: 1:1 (identical) to 21:1 (black on white)
```

You don't calculate this yourself. WebAIM's checker does it instantly.

The three thresholds to know

```
WCAG 1.4.3 (Level AA):
  Normal text (under 18pt or 14pt bold):  4.5:1
  Large text (18pt+ or 14pt bold+):       3:1

WCAG 1.4.6 (Level AAA, enhanced):
  Normal text:  7:1
  Large text:   4.5:1

WCAG 1.4.11 (Non-text contrast):
  Button borders, input outlines, icons:  3:1
```

Target AA. Aim for AAA on body text if you can reach it without sacrificing hierarchy.

(WebAIM, 2021)

The most common failure: mid-range color as text

Step 500 of a neutral ramp against a white or near-white background typically achieves 3.5:1, passing AA for large text, failing for body text.

```
Neutral/500: hsl(220, 9%, 50%)
on Neutral/50: hsl(220, 9%, 97%)
Ratio: ~3.5:1  -- FAILS AA for normal text
```

Fix: move text to step 600 or 700. Step 600 on Neutral/50 typically clears 4.5:1.

The accent color trap

Saturated colors at step 500 often fail AA for text.

- Blue-family colors (hue 200-260) at step 600-700 usually pass AA
- Red and orange at the same step often fail because their luminance is lower than blue at equivalent HSL lightness

Rule: always test accent colors as text. Never assume a mid-range step passes.

The fix is to darken the accent one or two steps, not to desaturate it.

Semantic color tokens

Two layers: primitive and semantic

Primitive layer: names what a color is.

```
Neutral/700, Accent/500, Red/600
```

Semantic layer: names what a color does.

```
text-primary, accent-default, surface-base, danger-default
```

Each semantic token references exactly one primitive value.

When you switch modes (light to dark), only the semantic assignments change. The components don't change.

A minimal semantic set

Eight tokens cover most product surfaces:

```
surface-base        -- page or panel background
surface-elevated    -- card or modal background
text-primary        -- headings, high-emphasis text
text-secondary      -- captions, labels, supporting text
accent-default      -- buttons, links, primary action
accent-subtle       -- tinted backgrounds, badge fills
border-default      -- dividers, input outlines
danger-default      -- errors, destructive actions
```

Start here. Add tokens only when you find a real gap.

Why semantic tokens matter

Without them:

- Adding dark mode requires editing every component individually
- A brand color change requires finding every use of that hex value
- There is no single source of truth for "what color is this component's background"

With them:

- Dark mode is a new set of eight assignments
- A brand change updates one primitive value, propagates everywhere
- Every component fill refers to a named role, not a number

(Khamatov, Smashing Magazine, 2023; Miao, Figma, 2022)

Figma variables: how to implement this

In Figma, create two variable collections:

1. **Primitives**, one mode (Light), nine steps per ramp, named `Neutral/100` through `Neutral/900`
2. **Semantic**, two modes (Light, Dark), named by function (`surface-base`, `text-primary`, etc.), each value referencing a primitive

The mode switch on the Semantic collection is what powers one-click dark mode.

(Figma, Tokens, Variables, and Styles, 2023)

Dark mode as a parallel system

Why inversion fails

Naively inverting light-mode values produces wrong results:

- `Neutral/50` (light surface) inverted becomes `Neutral/950`, correct
- `Neutral/600` (secondary text in light) inverted becomes `Neutral/400`, FAILS contrast on dark surface
- Accent colors at step 500 may not pass against a dark background

Inversion is arithmetic. Dark mode requires judgment.

The correct approach

For each semantic token, ask: given a dark surface, what primitive value serves this function well?

```
Light mode                Dark mode
surface-base:  N/50   -> N/950 (correct: dark base)
text-primary:  N/900  -> N/50  (correct: light text)
text-secondary:N/600  -> N/300 (NOT N/400: check contrast)
accent-default:A/600  -> A/400 (lighter for dark bg)
border-default:N/200  -> N/700 (subtle on dark surface)
```

Notice: `text-secondary` does not mirror symmetrically. It moves to N/300, not N/400.

The text-secondary failure mode

This is the most common dark-mode contrast failure in production products.

```
Light mode: N/600 on N/50
  Ratio: ~5.9:1   -- passes AA

Dark mode (naive): N/600 on N/950
  Ratio: ~2.4:1   -- FAILS AA (badly)

Dark mode (corrected): N/300 on N/950
  Ratio: ~7.1:1   -- passes AAA
```

Test every semantic token explicitly in dark mode. The numbers change dramatically.

(Miao, Figma Engineering Blog, 2022)

Dark mode accents: go lighter

An accent color at step 600 on a white surface:

- `Accent/600` on `Neutral/50`: ~6.8:1, passes AA

The same accent on a dark surface:

- `Accent/600` on `Neutral/950`: ~3.0:1, fails AA for normal text

Fix: move to Accent/400 or Accent/300 on dark surfaces. Test until you clear 4.5:1.

The Figma workflow for dark mode

1. Open the Semantic variable collection
2. Switch to the Dark mode column
3. For each token, click the value and reassign it to the appropriate dark primitive
4. Apply the Dark mode to a frame using the Variable Mode panel
5. Test every contrast pair in dark mode explicitly

The structure is identical to light mode. Only the values change.

Take-aways

1. Build ramps using the Tailwind 50-950 structure. Each step has a job; name the job before assigning the color.
2. Target WCAG AA (4.5:1 for normal text, 3:1 for large text). Mid-range palette steps almost always fail for body text, move text to step 600 or darker.
3. Semantic tokens separate what a color is from what it does. With a primitive and semantic layer, dark mode is eight new assignments, not a redesign.
4. Dark mode is not inversion. Derive each dark assignment from its function. Text-secondary is the most common failure: it needs to move two or three steps lighter on a dark surface.

> "We had 5,100 color variables that we needed to support both light and dark mode for... The infrastructure work we did for Dark Mode Week has helped us add new features faster than ever."

Shirley Miao, Figma Engineering Blog (2022)

What's next

- **Section (this week):** build your product's full palette as Figma variables and apply it to your starter kit
- **Reading:** Week 4 reading, Color, palettes, contrast, and dark mode
- **Week 5:** Spacing, the 8-point grid, and responsive layout in Figma

No assignment due this week.